

Vellore Institute of Technology

Thermal Management System for a Battery Pack

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The EV Battery Challenge

What Are Electric Vehicles?

An electric vehicle (EV) is a motor vehicle whose propulsion is powered fully or mostly by electricity, using one or more electric motors for movement instead of an internal combustion engine.

Key Points

Electric vehicles depend on lithium-ion batteries.

Problem 1: Batteries are highly sensitive to temperature.

Problem 2: High temperatures lead to faster degradation and safety risks, including fire.

Problem 3: Low temperatures reduce performance and range.

Crucial issue: Uneven temperatures cause cells to age differently, which shortens the overall lifespan of the battery.

Solution: An efficient battery thermal management system (BTMS) is needed.

Project Goals

Primary Objectives

Maintain optimal temperature: Keep the battery within its ideal range of 20 °C to 40 °C.

Design and model: Develop a conceptual design for a cooling system, using either liquid cooling or phase change materials (PCMs).

Secondary Objectives

Improve uniformity: Minimize temperature differences across the battery pack.

Evaluate efficiency: Design the system so that it uses minimal energy.

The work applies to battery electric vehicles, plug-in hybrid electric vehicles and hybrid electric vehicles.

Methodology

The project follows the process described below.

1. Research: A literature review on BTMS.

2. Design: Theoretical modelling of heat generation and heat transfer.

3. Simulation: PyCharm is used to run the simulation and show the data extraction.

4. Analysis: The data is collected and analysed to evaluate performance.

5. Conclusion: The findings are summarised and conclusions are drawn.

Literature Review

The following studies were reviewed for this project.

1. A Comprehensive Review of Thermal Management Challenges and Safety Considerations in Lithium-Ion Batteries for Electric Vehicles
2. Review of Battery Thermal Management Systems in Electric Vehicles
3. A Comprehensive Review of Thermal Management Methods and Ideal System Design for Improved Electric Vehicle Battery Pack Performance and Safety
4. Advances in the Battery Thermal Management Systems of Electric Vehicles for Thermal Runaway Prevention and Suppression
5. Phase Change Materials Application in Battery Thermal Management System: A Review

Sample Code

Sample code used for the project is given below.

Parallel Assembly Type 1

Simscape Battery Component Description and Cell Data Comparison

Overview

This document describes the Simscape component named Parallel Assembly Type 1. It is a parallel assembly block of battery cells, generated with the Simscape Battery tools. The model resolution is lumped, which means the ten cells in the assembly are represented by a single cell model together with lumping adapters that scale its electrical and thermal behaviour. The component was generated on 12 November 2025 at 23:33:44 using MATLAB version 25.2 and Simscape Battery version 25.2.

Cell Lookup Table Parameters

The cell is described by lookup tables that depend on the state of charge and on the temperature. The state of charge values are 0, 0.1, 0.25, 0.5, 0.75, 0.9 and 1. The temperature values are 278 K, 293 K and 313 K.

The open-circuit voltage table gives the following values.

At a state of charge of 0, the open-circuit voltage is 3.49 V at 278 K, 3.50 V at 293 K and 3.51 V at 313 K.

At a state of charge of 0.1, the open-circuit voltage is 3.55 V at 278 K, 3.57 V at 293 K and 3.56 V at 313 K.

At a state of charge of 0.25, the open-circuit voltage is 3.62 V at 278 K, 3.63 V at 293 K and 3.64 V at 313 K.

At a state of charge of 0.5, the open-circuit voltage is 3.71 V at 278 K, 3.71 V at 293 K and 3.72 V at 313 K.

At a state of charge of 0.75, the open-circuit voltage is 3.91 V at 278 K, 3.93 V at 293 K and 3.94 V at 313 K.

At a state of charge of 0.9, the open-circuit voltage is 4.07 V at 278 K, 4.08 V at 293 K and 4.08 V at 313 K.

At a state of charge of 1, the open-circuit voltage is 4.19 V at 278 K, 4.19 V at 293 K and 4.19 V at 313 K.

The terminal voltage operating range runs from 0 V up to infinity, so no upper or lower limit is imposed on the terminal voltage.

The terminal resistance table gives the following values.

At a state of charge of 0, the terminal resistance is 0.0117 ohm at 278 K, 0.0085 ohm at 293 K and 0.009 ohm at 313 K.

At a state of charge of 0.1, the terminal resistance is 0.011 ohm at 278 K, 0.0085 ohm at 293 K and 0.009 ohm at 313 K.

At a state of charge of 0.25, the terminal resistance is 0.0114 ohm at 278 K, 0.0087 ohm at 293 K and 0.0092 ohm at 313 K.

At a state of charge of 0.5, the terminal resistance is 0.0107 ohm at 278 K, 0.0082 ohm at 293 K and 0.0088 ohm at 313 K.

At a state of charge of 0.75, the terminal resistance is 0.0107 ohm at 278 K, 0.0083 ohm at 293 K and 0.0091 ohm at 313 K.

At a state of charge of 0.9, the terminal resistance is 0.0113 ohm at 278 K, 0.0085 ohm at 293 K and 0.0089 ohm at 313 K.

At a state of charge of 1, the terminal resistance is 0.0116 ohm at 278 K, 0.0085 ohm at 293 K and 0.0089 ohm at 313 K.

The cell capacity is 27 ampere hours. Values that fall outside the tables are handled by taking the nearest table value. The thermal mass of the cell is 100 joules per kelvin.

Cell Balancing Parameters

Cell balancing is carried out by a switch in series with a shunt resistor. The switch has a closed resistance of 0.01 ohm and an open conductance of 0.00000001 per ohm. It operates when its control signal passes a threshold of 0.5. The shunt resistance is 50 ohm.

Thermal Parameters

The cell level coolant thermal path resistance is 1.2 kelvin per watt, and the cell level ambient thermal path resistance is 25 kelvin per watt. Because ten cells are connected in parallel, these values are scaled for the whole assembly. The scaled coolant resistance is the coolant resistance divided by the number of cells in parallel, which gives 0.12 kelvin per watt. The scaled ambient resistance is the ambient resistance divided by the number of cells in parallel, and it is then multiplied by the number of cell thermal node elements and divided by the number of rows in the ambient resistance value, so that the thermal behaviour is shared correctly between the thermal nodes.

Internal Parameters

The number of batteries in parallel is 10. The number of cell thermal node elements is taken from the number of rows in the thermal port of the cell model. These values are used only inside the component and cannot be changed from outside.

Input

The component has one input, called enable cell balancing, with a default value of 0. When this input rises above the balancing threshold, the balancing switch closes and the shunt resistor is connected across the assembly.

Nodes

The component has four connection nodes. The positive and negative nodes are electrical terminals. The top external coolant node and the ambient node are thermal terminals. The positive terminal is drawn on the top side of the block, the negative terminal on the bottom side, and the coolant terminal on the top side.

Variables

The component tracks the cell current, which is positive when flowing into the cell and starts at 0 A, and the cell terminal voltage, which starts at 0 V. It also tracks the cell state of charge, which starts at 1 and is given high priority, the number of cell discharge cycles, which starts at 0 and is given high priority, and the cell temperature, which starts at 298.15 K and is given high priority. In addition, the parallel assembly voltage starts at 0 V and the parallel assembly state of charge starts at 1, and both have no priority.

Intermediate Quantity

The power dissipated by the cells other than the modelled one is calculated by multiplying the power dissipated by the single cell model by the number of cells in parallel minus one.

Equations

Each of the variables listed above is required to be a single value. The parallel assembly voltage is equal to the voltage across the positive and negative terminals of the cell. The parallel assembly state of charge is equal to the state of charge of the cell. The current, voltage, state of charge, number of discharge cycles and temperature of the cell model are each set equal to the matching cell variables described above.

Components

The main component is a table battery cell. It uses the lookup tables, capacity, extrapolation setting and thermal mass described earlier. Its state of charge port is disabled and temperature dependence is switched on. The ageing of the open-circuit voltage is set to the open-circuit voltage option and ageing is modelled by equation, while capacity ageing and resistance ageing are disabled. Current directionality, dynamic behaviour, capacity fade and self-discharge leakage are all switched off. The thermal port is modelled, and the current, voltage, state of charge, cycle count and temperature of the cell are all given no priority.

The remaining components are the balancing switch, the balancing resistor and a coolant thermal resistance at the top of the assembly, which uses the scaled coolant resistance. There are also electrical lumping adapters on the positive and negative sides, each set for ten cells in parallel, and a thermal lumping adapter for every cell thermal node element. Finally, one ambient thermal resistor is created for each thermal node element, and each of them uses the corresponding scaled ambient resistance value.

Connections

The thermal nodes of the cell are connected to the thermal lumping adapters. The negative terminal of the assembly is connected to the negative lumping adapter, which is connected to the negative terminal of the cell. In the same way, the positive terminal of the assembly is connected to the positive lumping adapter, which is connected to the positive terminal of the cell.

The balancing switch and the balancing resistor are connected in series between the lumped positive node and the lumped negative node. The enable cell balancing input drives the control terminal of the switch. The lumped thermal node of the first thermal adapter is connected to the

coolant resistance, whose other end goes to the top external coolant node. The lumped thermal node of every thermal adapter is also connected to its own ambient resistor, and the other end of each ambient resistor is connected to the ambient node.

User Interface Layout and Icon

The block dialog is organised into three groups. The Main group holds the state of charge values, temperature values, open-circuit voltage table, terminal voltage range, terminal resistance table, cell capacity and extrapolation method. The Thermal group holds the thermal mass, coolant resistance and ambient resistance. The Cell Balancing group holds the closed resistance, open conductance, threshold and shunt resistance of the balancing circuit. The block uses the icon file parallelAssembly.svg.

Comparison of Cell Data at Different Temperatures

The two figures below compare the open-circuit voltage and the terminal resistance of the cell at the three temperatures in the lookup tables.

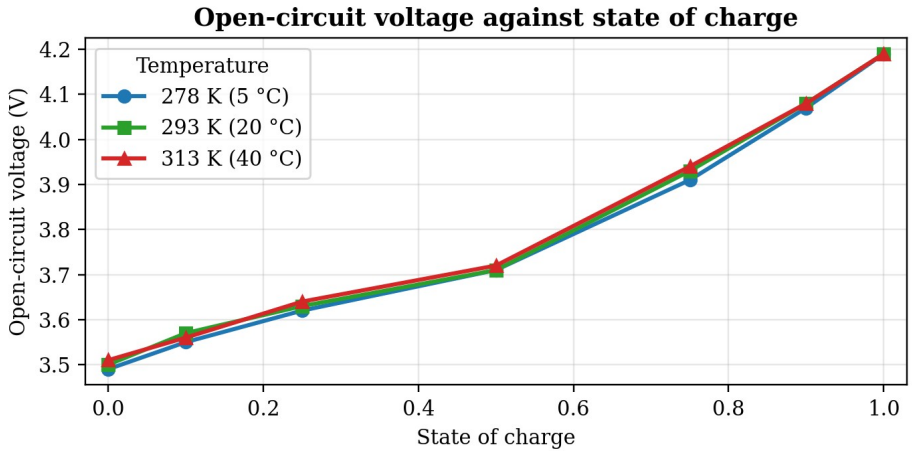


Figure 1 shows that the open-circuit voltage rises from about 3.5 V at empty to 4.19 V at full charge, and that the three temperature curves almost overlap. The largest difference between temperatures at any state of charge is only 0.03 V, so temperature has little effect on the open-circuit voltage.

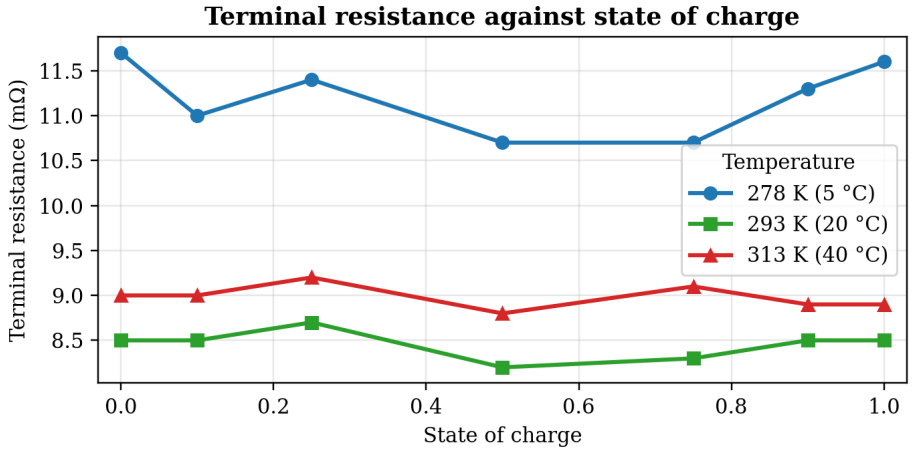


Figure 2 shows that temperature has a much stronger effect on the terminal resistance. At 278 K the resistance is between 10.7 and 11.7 milliohm. At 293 K it is between 8.2 and 8.7 milliohm, and at 313 K it is between 8.8 and 9.2 milliohm. The lowest resistance is therefore found at 293 K and not at the highest temperature, and the cold condition at 278 K gives the highest resistance and so the largest voltage drop under load.

MATLAB Simscape Code

Source file: ParallelAssemblyType1.ssc

```
component ParallelAssemblyType1
    % ParallelAssemblyType1:2
    % Parallel Assembly Block
    %    Model resolution: Lumped
    %    Number of cells: 10
    %
    %    MATLAB version: 25.2
    %    Simscape Battery version: 25.2
    %    Simscape code generated on: 12-Nov-2025 23:33:44

    parameters
        SOC_vecCell = {[0, .1, .25, .5, .75, .9, 1],'1'}; % Vector of state-of-charge values, SOC
        T_vecCell = {[278, 293, 313],'K'}; % Vector of temperatures, T
        V0_matCell = {[3.49, 3.5, 3.51; 3.55, 3.57, 3.56; 3.62, 3.63, 3.64; 3.71, 3.71, 3.72; 3.91, 3.93, 3.94; 4.07, 4.08, 4.08; 4.19, 4.19, 4.19],'V'}; % Open-circuit voltage, V0(SOC,T)
        V_rangeCell = {[0, inf],'V'}; % Terminal voltage operating range [Min Max]
        R0_matCell = {[.0117, .0085, .009; .011, .0085, .009; .0114, .0087, .0092; .0107, .0082, .0088; .0107, .0083, .0091; .0113, .0085, .0089; .0116, .0085, .0089],'Ohm'}; % Terminal resistance, R0(SOC,T)
        AHCell = {27,'A*hr'}; % Cell capacity, AH
        extrapolation_optionCell = {simscape.enum.extrapolation.nearest,'1'}; % Extrapolation method for all tables
        thermal_massCell = {100,'J/K'}; % Thermal mass
        CellBalancingClosedResistance = {0.01,'Ohm'}; % Cell balancing switch closed resistance
        CellBalancingOpenConductance = {1e-8,'1/Ohm'}; % Cell balancing switch open conductance
        CellBalancingThreshold = {0.5,'1'}; % Cell balancing switch operation threshold
        CellBalancingResistance = {50,'Ohm'}; % Cell balancing shunt resistance
        CoolantResistance = {1.2,'K/W'}; % Cell level coolant thermal path resistance
        AmbientResistance = {25,'K/W'}; % Cell level ambient thermal path resistance
    end

    parameters(ExternalAccess=none)
        P = 10; % Batteries in Parallel
        NumCellThermalNodeElements = size(TBCell(1).H,1); % NumCellThermalNodeElements
    end

    parameters(Access=private)
        AmbientResistanceScaled = AmbientResistance .* ones(NumCellThermalNodeElements,1) /P * NumCellThermalNodeElements / size(AmbientResistance,1); % Scaled cell level ambient thermal path resistance
        CoolantResistanceScaled = CoolantResistance .* ones(1,1) /P; % Scaled cell level coolant thermal path resistance
    end

    inputs
        enableCellBalancing = {0,'1'}; % CB
    end

    nodes
        p = foundation.electrical.electrical; % +
        n = foundation.electrical.electrical; % -
        TopExtClnt = foundation.thermal.thermal; % CPT
        AmbH = foundation.thermal.thermal; % AmbH
    end

    variables
        iCell = {0,'A'}; % Cell current (positive in)
        vCell = {0,'V'}; % Cell terminal voltage
```

```
52          socCell = {value={1,'1'},priority=priority.high}; % Cell state of charge
53          numCyclesCell = {value={0,'1'},priority=priority.high}; % Cell discharge cycles
54          temperatureCell = {value={298.15,'K'},priority=priority.high}; % Cell temperature
55          vParallelAssembly = {value={0,'V'},priority=priority.none}; % Parallel Assembly Voltag
                e
56          socParallelAssembly = {value={1,'1'},priority=priority.none}; % Parallel Assembly stat
                e of charge
57      end
58
59      intermediates
60          power_dissipated = TBCell.power_dissipated*(P-1); % Power dissipated by P-1 cells
61      end
62
63      equations
64          assert(length(iCell) == 1);
65          assert(length(vCell) == 1);
66          assert(length(socCell) == 1);
67          assert(length(numCyclesCell) == 1);
68          assert(length(temperatureCell) == 1);
69          assert(length(vParallelAssembly) == 1);
70          assert(length(socParallelAssembly) == 1);
71          vParallelAssembly == TBCell(1).p.v - TBCell(1).n.v;
72          socParallelAssembly == TBCell.stateOfCharge;
73          TBCell.i == iCell;
74          TBCell.v == vCell;
75          TBCell.stateOfCharge == socCell;
76          TBCell.num_cycles == numCyclesCell;
77          TBCell.cell_temperature == temperatureCell;
78      end
79
80      components(ExternalAccess=observe)
81          TBCell = batteryecm.table_battery(SOC_vec = SOC_vecCell,...
82              T_vec = T_vecCell,V0_mat = V0_matCell,V_range = V_rangeCell,R0_mat = R0_matCell,..
                    .
83              AH = AHCell,extrapolation_option = extrapolation_optionCell,...
84              thermal_mass = thermal_massCell,SOC_port = simscape.enum.tablebattery.enable.no,..
                    .
85              T_dependence = simscape.enum.tablebattery.temperature_dependence.yes,...
86              prm_age_OCV = simscape.enum.tablebattery.prm_age_OCV.OCV,...
87              prm_age_capacity = simscape.enum.tablebattery.prm_age.disabled,...
88              prm_age_modeling = simscape.enum.tablebattery.prm_age_modeling.equation,...
89              prm_age_resistance = simscape.enum.tablebattery.prm_age.disabled,...
90              prm_dir = simscape.enum.tablebattery.prm_dir.noCurrentDirectionality,...
91              prm_dyn = simscape.enum.tablebattery.prm_dyn.off,prm_fade = simscape.enum.tablebat
                    tery.prm_fade.disabled,...
92              prm_leak = simscape.enum.tablebattery.prm_leak.disabled,...
93              thermal_port = simscape.enum.thermaleffects.model,i.priority = priority.none,...
94              v.priority = priority.none,stateOfCharge.priority = priority.none,...
95              num_cycles.priority = priority.none,cell_temperature.priority = priority.none);
96          balancingSwitch = foundation.electrical.elements.controlled_switch(R_closed = CellBala
                ncingClosedResistance,...
97              G_open = CellBalancingOpenConductance,Threshold = CellBalancingThreshold);
98          balancingResistor = foundation.electrical.elements.resistor(R = CellBalancingResistanc
                e);
99          CoolantResistanceTop = foundation.thermal.elements.resistance(resistance = CoolantResi
                stanceScaled(1));
100     end
101
102     components(Access=private)
103         PositiveAdapter = TBBATTERYLumpingAdapters.ElectricalLumpingAdapter(CellsInParallel =
                P);
104         NegativeAdapter = TBBATTERYLumpingAdapters.ElectricalLumpingAdapter(CellsInParallel =
                P);
105         ThermalAdapter = repmat(TBBATTERYLumpingAdapters.ThermalLumpingAdapter(CellsInParallel
                 = P),NumCellThermalNodeElements,1);
106     end
```

```
107
108     connections
109         connect([TBCell.H],[ThermalAdapter.CellNode]');
110         connect(n,NegativeAdapter.LumpedNode);
111         connect(NegativeAdapter.CellNode,TBCell.n);
112         connect(p,PositiveAdapter.LumpedNode);
113         connect(PositiveAdapter.CellNode,TBCell.p);
114         connect(PositiveAdapter.LumpedNode,balancingSwitch.p);
115         connect(balancingSwitch.n,balancingResistor.p);
116         connect(balancingResistor.n,NegativeAdapter.LumpedNode);
117         connect(enableCellBalancing,balancingSwitch.vT);
118         connect([ThermalAdapter(1).LumpedNode]',CoolantResistanceTop.A);
119         connect(TopExtClnt,CoolantResistanceTop.B);
120         connect([ThermalAdapter.LumpedNode],[AmbientResistor.A]);
121         connect(AmbientResistor(:).B,AmbH);
122     end
123
124     annotations
125         [p] : Side=top;
126         [n] : Side=bottom;
127         [TopExtClnt] : Side=top;
128         UILayout = [UIGroup("Main",SOC_vecCell,T_vecCell,V0_matCell,V_rangeCell,R0_matCell,AHC
                ell,extrapolation_optionCell),...
129             UIGroup("Thermal",thermal_massCell,CoolantResistance,AmbientResistance),...
130             UIGroup("Cell Balancing",CellBalancingClosedResistance,CellBalancingOpenConductanc
                    e,CellBalancingThreshold,CellBalancingResistance)];
131         Icon = 'parallelAssembly.svg';
132     end
133
134     for thermalNodeElementIdx = 1:NumCellThermalNodeElements
135         for modelIdx = 1:1
136             components(ExternalAccess=observe)
137                 AmbientResistor(thermalNodeElementIdx,modelIdx) = foundation.thermal.elements.
                        resistance(...
138                     resistance = AmbientResistanceScaled(thermalNodeElementIdx,modelIdx));
139             end
140         end
141     end
142 end
```